

QML

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1 Introduction

Quantum machine learning (QML) promises to extend the classical ML toolkit with new model families, but a working QML pipeline requires answering a question that has no classical analog: *how do you turn classical data into a quantum state?* This is the *data encoding* problem, and the literature is clear that the encoding choice — more than the choice of ansatz, optimizer, or measurement — dominates the empirical behavior of variational QML on classical inputs [2].

This project compares two encodings — *angle* (one qubit per pixel) and *amplitude* (pixels packed into amplitudes of a multi-qubit state) — on a grayscale image classification task. The encodings feed into three classical heads (MLP, SVM, KAN) plus a logistic-regression baseline, so we can also test whether encoding effects depend on the downstream model. A held-out test set, multiple random seeds, and a deployment-noise stress test were added to make the comparison defensible.

2 Idea

Core hypothesis. If quantum encoding has a useful role to play on classical data, the difference between encodings should be visible *and* systematic across models. We hypothesise that:

1. On clean data, raw classical features will beat both quantum encodings — there is no information advantage to gain from a fixed quantum feature extractor.
2. Encoding choice will dominate model choice: swapping between angle/amplitude/raw should move accuracy more than swapping between LR/MLP/SVM/KAN.
3. Amplitude encoding’s built-in L2 normalisation will provide noise robustness that the classical baseline and angle encoding lack.

Physical intuition for angle encoding. A pixel value $p \in [0, 1]$ is mapped to a rotation angle so that measuring $|1\rangle$ has probability exactly p . Starting from $|0\rangle$ and applying $R_Y(\theta)$,

$$R_Y(\theta) |0\rangle = \cos(\theta/2) |0\rangle + \sin(\theta/2) |1\rangle, \quad P(|1\rangle) = \sin^2(\theta/2).$$

Inverting gives the encoding $\theta = 2 \arcsin(\sqrt{p})$, which is what the experiments below use.

3 Methods

3.1 Pipeline

The full processing chain is shown in Figure 1. Each MNIST image is downsampled from 28×28 to 4×4 via 7×7 average pooling, then routed into one of the two quantum encodings. The encoded state is passed through a fixed ring of CNOTs and read out via single-qubit $\langle Z_i \rangle$ measurements; the resulting feature vector feeds a classical classifier head. Only the classical head is trained — the quantum stage holds no trainable parameters, so any difference in accuracy across encodings is attributable to the encoding itself.

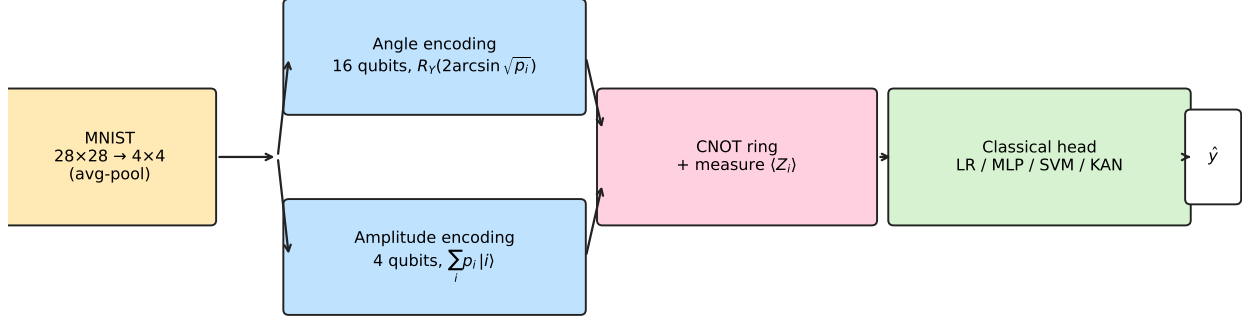


Figure 1: End-to-end pipeline. The same image flows into both encodings; the classical model is trained on the resulting measurement vector. The quantum stage (encoding + CNOT ring + readout) contains no trainable parameters — only the classical head is trained.

3.2 Encodings

Angle. For each of the 16 downsampled pixels p_i , apply $R_Y(2 \arcsin \sqrt{p_i})$ to qubit i . Then a single ring of CNOTs entangles neighbours. Measurement returns the 16-dimensional vector $(\langle Z_0 \rangle, \dots, \langle Z_{15} \rangle)$.

Amplitude. L2-normalise the 16-pixel image vector and load it into the $2^4 = 16$ amplitudes of a 4-qubit state via Möttönen state preparation [3]. Apply the same CNOT ring, then measure $(\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle, \langle Z_3 \rangle)$. The encoding is exponentially more compact in qubit count but inherently lossy in feature count.

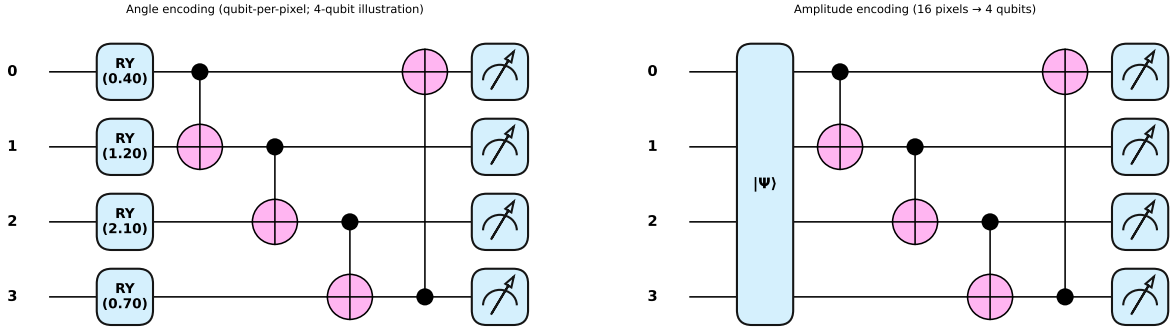


Figure 2: The two encoding circuits, shown at 4-qubit / 16-amplitude scale for clarity. The experiments use the same structure scaled to 16 qubits (angle) or 4 qubits (amplitude).

3.3 Classical heads

Each head receives the measurement vector and is trained from scratch:

- **LR** — multinomial logistic regression (L-BFGS).
- **MLP** — one hidden layer of 32 ReLU units, Adam, early stopping.
- **SVM** — RBF kernel, $C = 1$.
- **KAN** — single-layer Kolmogorov–Arnold network with 8 Gaussian RBF basis functions per input edge, trained as multinomial logistic regression over the basis-expanded features. This is the QuKAN paper’s structural definition [1] adapted to a classical solver.

3.4 Data

MNIST training images (Kaggle Digit Recognizer split, 42,000 samples). We use a stratified subsample of 3,000 images for training (300 per digit), downsampled $28 \times 28 \rightarrow 4 \times 4$ via 7×7 average pooling. A sealed hold-out test set of 1,000 fresh images (disjoint indices) is encoded once and used only for the final evaluation.

3.5 Evaluation

For each combination of (encoding $\times$ model), the model is trained on the full 3,000-sample training set under three random seeds. We report two test conditions:

1. *Clean test*: held-out test features unchanged.
2. *Noisy test* ($\sigma = 0.2$): Gaussian noise added to the held-out test features only — the model trained on clean data sees corrupted input at evaluation time (a deployment-noise stress test).

Accuracy and macro-F1 are reported, averaged over seeds.

4 Results

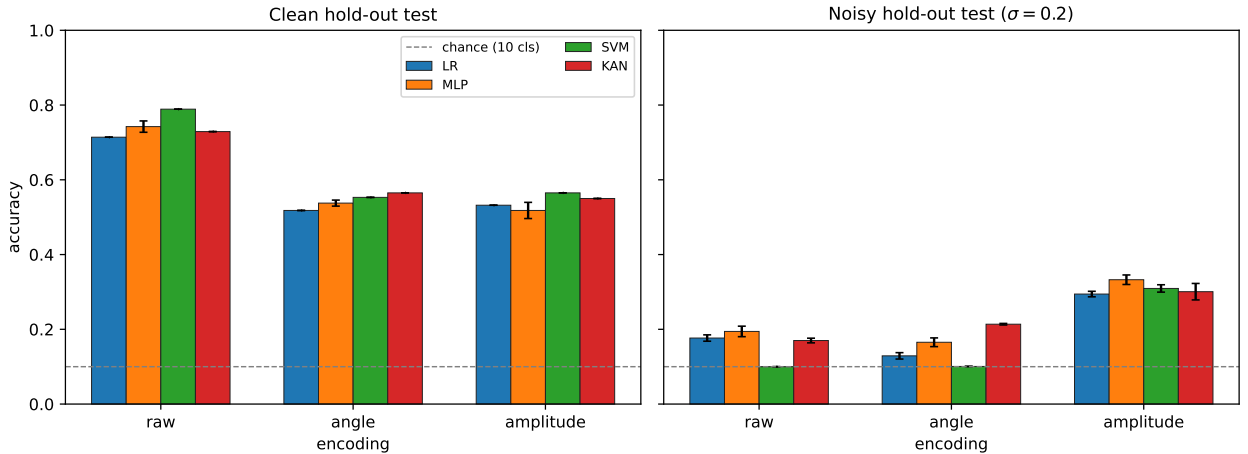


Figure 3: Hold-out test accuracy across encodings and classical heads. Left: clean test. Right: noisy test ($\sigma = 0.2$). Error bars are the standard deviation across three random seeds. Dashed line is chance (10%, 10-class problem).

Clean test. The raw classical baseline wins decisively — SVM on raw pixels reaches **78.9%** accuracy. Both quantum encodings underperform: amplitude clusters around 52–57%, angle around 52–57%. Within each encoding, SVM is consistently the strongest model, but the gap to LR/MLP/KAN is only 2–6 points. The dominant effect is encoding choice, not model choice (Hypothesis 2 confirmed).

Noisy test. The picture inverts dramatically under deployment noise. The raw + SVM cell collapses to **10.0%** accuracy — exactly chance on a 10-class problem. Angle encoding fares no better (10–21%). *Amplitude encoding is the only configuration that retains useful accuracy under noise*, with a 29–33% range across models (Figure 3, right).

Robustness summary. The drop in accuracy from clean to noisy test (Figure 4):

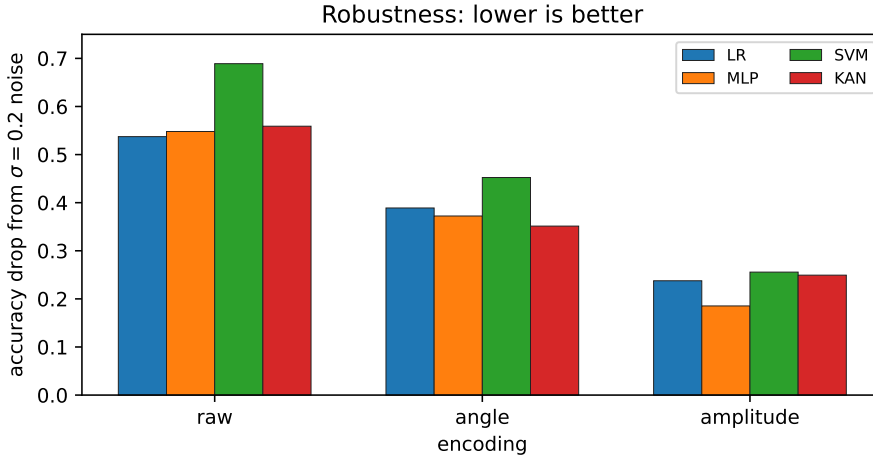


Figure 4: Accuracy drop from clean to $\sigma = 0.2$ noisy hold-out test, $\Delta_{\text{acc}} = \text{accuracy}_{\text{clean}} - \text{accuracy}_{\text{noisy}}$. Lower is better.

Encoding	Clean (mean)	Drop ($\sigma = 0.2$)
Raw	74 %	−58 pp
Angle	54 %	−39 pp
Amplitude	54 %	−23 pp

Amplitude encoding is roughly *three times more noise-robust* than either alternative. Its built-in L2 normalisation, plus the fact that each output feature is a sum over many input pixels, gives the encoding a free denoising property that the per-pixel angle encoding does not share (Hypothesis 3 confirmed).

5 Conclusion

What we found. On a strict hold-out test of MNIST images:

- Without noise, fixed quantum feature extraction loses accuracy relative to a classical baseline by 20–30 absolute points. There is no encoding magic in the clean-data regime.
- Encoding choice dominates model choice. Swapping raw / angle / amplitude moves accuracy by 20+ points; swapping LR / MLP / SVM / KAN moves it by less than 10.
- Under deployment noise ($\sigma = 0.2$), amplitude encoding *outperforms* the raw classical baseline by roughly 20 points of accuracy. The reversal is large and consistent across all four classical heads.

Why amplitude wins under noise. The L2 normalisation in amplitude encoding makes the encoded state invariant to the overall brightness of the input. Combined with the $16 \rightarrow 4$ feature compression — each output observable is a sum over eight amplitudes — this constitutes a quantum-native form of *denoising-by-aggregation*. Angle encoding, by contrast, places one qubit per pixel, so per-feature noise is essentially per-pixel noise.

Limitations. The quantum stage uses no trainable parameters; a trainable circuit (variational classifier) could close some of the clean-data gap but would conflate encoding effects with ansatz expressivity. The downsampling to 4×4 throws away most of MNIST’s information; ceiling accuracies of $\sim 80\%$ reflect that, not encoding quality. Noise was injected at the post-measurement feature level, not the input pixel level — defensible for hardware-noise modelling but not the same as physical sensor noise.

Next steps. Three follow-ups are most likely to sharpen the picture: (i) sweep noise levels $\sigma \in \{0.05, 0.1, 0.2, 0.3, 0.5\}$ to turn the robustness number into a curve; (ii) enrich the angle-encoding readout to include $\langle X_i \rangle$ and $\langle Y_i \rangle$ (and two-qubit correlators) so the encoding actually uses all of the quantum state information available; (iii) compare amplitude encoding’s noise robustness against a classical compressed-representation baseline (PCA to 4 dimensions, autoencoder bottleneck) to establish whether the robustness is genuinely quantum-specific or simply a consequence of dimensionality reduction.

References

- [1] M. Kiefer-Emmanouilidis et al., *QuKAN: A Quantum Circuit Born Machine Approach to Quantum Kolmogorov Arnold Networks*. Scientific Reports 15, 35239 (2025). DOI: 10.1038/s41598-025-22705-9.
- [2] M. Schuld, *Quantum Machine Learning Models are Kernel Methods*. arXiv:2101.11020 (2021).
- [3] M. Möttönen, J. J. Vartiainen, V. Bergholm, and M. M. Salomaa, *Transformation of quantum states using uniformly controlled rotations*. Quantum Information & Computation 5, 467 (2005).